

Advanced Calculus And Complex Analysis

Sem 2

Watermarked study resource

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STATUS: ok

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Reg. No.
Max. Marks: 75Time: 3 hours
Marks BL CO PO

1 1
1 I 112.
. i i i i
i i i i
i 2 2 2
2 i - 2 j + k
C) i + j + k
i i 2 16.
2 11 17.
i i 2 18.
i i 3 1
20NF221MAB102T
PART - A (20 x 1 = 20 Marks)
Answer ALL Questions

B)3
D)2
B)2
D)6
B)1
D)3
B)0
D)3
B)1/s

D)1

B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, NOVEMBER 2024

Second Semester

21MAB102T - ADVANCED CALCULUS AND COMPLEX ANALYSIS

(For the candidates admitted from the academic year 2021-2022 to 2023 - 2024)

D) $2\sim i+2\sim 3+\sim k$

3

Note:

(i) Part - A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall

invigilator at the end of 40th minute.

(ii) Part - B & Part - C should be answered in answer booklet..

div r is

A) r

C)0

$\text{curl}(\text{grad}(f)) = ,$

A)1

C)2

The connection between a line integral and a double integral is known as

A) Gauss Divergence theorem B) Stoke's theorem

C) Green's theorem D) convolution theorem

$\int_0^2 \int_0^3 \int_0^1 dx dy dz$

$\int_0^1 \int_0^1 \int_0^1 dx dy dz$ is equal to

A)1

C)3

5. The unit vector normal to $\langle x, y, z \rangle =$

$\frac{1}{\sqrt{2}} \langle 1, 1, 0 \rangle$

3

1 • 1

+ $y^2 + z^2$ at $(1, -1, 2)$ is

$\sqrt{27} + 1$

9. $L(1) =$

A) s

C)0

Page 1 of 3

1 • Evaluate $\int_0^3 dx dy$

A)1

C)0

The curve $y^2 = 4x$ is a

A) parabola B) hyperbola

C) straight line D) ellipse

3. The area of the double integral in Cartesian co-ordinates is equal to

$\int_0^1 \int_0^1 dx dy$ B) $\int_0^1 \int_0^1 dx dy - k$ C) $\int_0^1 \int_0^1 x^2 dx dy$ D) $\int_0^1 \int_0^1 x^2 dx dy$

• A-5

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[Page 2]

11 1 3

1 1 3 111.

1 1 3 1

1 4 1 113. The function $f(z) = u + iv$ is analytic if

1 4 114.

i i 4 1

1 1 4 1

1 1 5 1

5 ' 1 1118.

i i19.

i • i

C) The real part of the Laurent's series

Marks BL CO PO

8 2

dydx.

8 3

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4

i-

PART - B (5 x 8 = 40 Marks)

Answer ALL Questions

B) magnification & rotation

D) translation

B)-1

D)k

B)UX~Vy

D)ux= -v.

2-zn=0

10. Find $L(\cos 2t)$

A)'?+2

C)7T4

$e(3+y+2)dx/dz$.

vx

1 2

A) rotation

C) magnification

17. What is the residue of $\cot(z)$ at $z = 0$?

A) 1

C)0

If $f(z) = (\sin z)/z$, then

A) $z = 0$ is a zero of $f(z)$ B) $z = 0$ is a simple pole

C) $z = 0$ is a pole of order 2 D) $z = 0$ is a removable singularity

Let $C : |z - a| = r$ be a circle, the $f(z)$ can be expanded as a Taylor series if

- A) $f(z)$ is not an analytic function within c B) $f(z)$ is an analytic function within c
 C) $f(z)$ is a defined function within c D) $f(z)$ is an analytic function outside c

20. The part $\sum_{n=0}^{\infty} a^n z^n$ consisting of positive integral powers of $(z - a)$ is called as

- A) The analytic part of the Laurent's series B) The principal part of the Laurent's series .

D) The imaginary part of the Laurent's series

ae^{-bt}

3×1

b-Evaluate $\int_0^1 x^2 dx$

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$f(x) = x^2 + y^2$

$\int_0^1 x dx$

- A) zero B) constant

- C) harmonic D) analytic

16. The transformation $w = z + c$ where c is a constant represents

An example of a function for which the Laplace transform does not exist is

A) $f(t) = t^2$ B) $f(t) = \tan t$

C) $f(t) = \sin t$ D) $f(t) = \cos t$

12. Using the initial value theorem, find the value of the function $f(t) =$

A) ab B) 0

C) a D) a^2

A) $u_x v_y, u_y v_x$

$\theta = -\nabla y' \cdot \nabla u - \nabla x$

If $u + iv$ is analytic, then the curves $u = C_1$ and $u = C_2$

A) are parallel B) intersect each other

C) cut orthogonally D) coincide

15. An analytic function with constant modulus is

21a. Evaluate by changing the order of integration

4a

(OR)

5×1

$\nabla u =$

$y \cdot \nabla u = v_x$

5×1

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[Page 3]

$8 \times 3 \times 2 \times 2$

$i \cdot 4 - \frac{1}{2} \frac{d^2}{ds^2} \sin a - 4$

find its scalar potential.

$8 \times 3 \times 2 \times 2$

$8 \times 4 \times 3 \times 2$

$8 \times 4 \times 4 \times 2$

8 4 2 3

8 4 4 2

8 . 3 2525 a. Evaluate

8 3 5 2

Marks BL CO PO

15 226. $i + I yz + 4 j - xzk$, where S is $y-z + 2$

21527. by using contour integration.

\$

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do

$13+5 \sin \theta$

- I-

(OR)

, where C is $|z| = 3$.

$(\sin t + \cos t z^2) dz$

$(z_1)^3 (z_2)$

PART - C (1 x 15 = 15 Marks)

Answer ANY ONE Question

+ 3rc2:2 k is irrotational and $y^2 \cos x + z^3$

^•Evaluate f_c

Verify Stoke's theorem for $I =$

the surface of the cube $x=0, x=2, y=0, y=2, z=0$ and $z=2$ above the xy -plane.

Evaluate

22 a. .Prove that

using convolution theorem.

(OR)

b-Using Laplace transform, solve $y = 1 - 2t$, given that $y=-1$ when $t=0$.

24 a. Determine the analytic function $f(z) = u + iv$ such that $u - v = e^x (\cos y - \sin y)$.

(OR)

b. Find the bilinear transformation that maps $Z = 0, 1, \infty$ into the points $W = i, 1, -i$.

25 a. Evaluate dz , where C is $|z|$

it^*

4 2

(OR)

b. Verify Green's theorem in the plane for $\oint_C (3x^2 - 8y^2)dx - (4y - 6xy)dy$, where C is the boundary of the region bounded by $x=0, y=0, x+y=1$.

23 a- $dL^1[(^$

4 5

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